



Cesar Augusto VALADES CRUZ, PhD

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Researcher with extensive experience in the development and operation of advanced optical microscopes, and in the processing and analysis of imaging data applied to cell biology and microbiology. His work integrates machine learning, large-scale data handling, high-content imaging, and 3D+time data visualization for the study of cellular processes. His research is organized around three main lines: (1) development and application of super-resolution microscopy, light-sheet microscopy, and nanoscale molecular orientation analysis; (2) 3D+time tracking of intracellular events; and (3) development of algorithms for large-scale biological image analysis.

PROFESSIONAL EXPERIENCE

Associate researcher — Algal Growth and Development TEAM of Prof. Zhang Cheng-Cai (02/2023 – Current)

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China

Research focuses on the study of the growth and differentiation of cyanobacterial cells using advanced microscopy and deep learning algorithms. Supervising Master & PhD Students.

Postdoctoral researcher — Team SERPICO Paris of Dr. C. Kervrann and Dr. J. Salamero (01/2022 – 01/2023)

French National Institute for Research in Digital Science and Technology (Inria) & CNRS - Curie Institute, Paris, France

Project: Polarization Microscopy for Imaging of Membrane Organization (PoMIMO). LabEx Cell(n)Scale collaborative project between Inria and Curie Institute. Implementation of polarization module in a STORM-TIRFM (PolarSTORM) system at Curie Institute. Developing of image analysis workflow of biomolecules tracking and estimating spatial high-resolution maps of molecular mobility.

Inria Starting Research Position — Team SERPICO Paris (04/2019 – 12/2021)

French National Institute for Research in Digital Science and Technology (Inria) & CNRS – Curie Institute, Paris, France

Research project: Acquisition, analysis, and visualization of 3D Dynamic cellular imaging of endocytosis/recycling mechanisms in the membrane during cell migration using machine learning. I work in novel machine learning methods of image processing able to detect the main regions of interest, and automatic quantification of molecular interactions and cell processes. In addition, I collaborate in the development of machine learning-driven navigation and interaction techniques for 3D+Time data enabling the analysis of localized intra-cellular events (endocytosis and exocytosis) and cell processes (migration, division, etc.).

Inria & startup Biotech Myriade, Paris, France

Industry collaborative project: Improvement of tracking and size estimation of virus and extracellular vesicles. Collaboration between SERPICO Team and the Biotech startup Myriade. I develop mathematical models and image processing tools to improve the tracking and size estimation of viruses and extracellular vesicles in VideoDrop.

Postdoctoral researcher — Team of Prof. Ludger JOHANNES, Curie Institute, Paris, France (01/2016 – 03/2019)

Project: Advanced cellular imaging of endocytosis. Development of 3D image processing and quantification methods to study different modes of endocytosis, using advanced high spatio-temporal resolution imaging and single particle tracking. In addition, I was also responsible of setting up a Lattice Light Sheet Microscope on the PICT-IBiSA imaging facility at Curie Institute, which is part of the France BioImaging National Infrastructure.

University Lecturer — Monterrey Institute of Technology and Higher Education, Mexico (01/2015 – 12/2015)

Teaching Physics, mathematics, and differential equations. Undergraduate courses: Physics I and II, Introduction to Physics, Physics for Design, and Differential Equations. Laboratory courses: Physics I and II, and Electricity and Magnetism.

EDUCATION

2010 – 2014 PhD, Photonics Engineering, Nanophotonics & Biophotonics

— Aix-Marseille University, Fresnel Institute, Marseille, France (Supervisor : Dr. Sophie BRASSELET)

— Polytechnic Univ. of Catalonia, Institute of Photonic Sciences, Barcelona, Spain (Supervisor : Dr. Pablo LOZA-ALVAREZ)

- Erasmus Mundus program. Thesis: Polarized Super-Resolution Fluorescence Microscopy. Implementation of a novel method of super resolution microscopy, in combination with a polarized detection to study molecular orientation behaviors, to report structural information at the single molecule and at nanometric spatial scale. Awarded Cum Laude.

- Responsible of developing image processing and quantification methods for super-resolution microscopy (dSTORM).

2008 – 2010 MSc, Biophotonics for Telecommunications and Biotechnologies — Ecole Normale Supérieure de Cachan, France & Complutense University of Madrid, Spain

- Erasmus Mundus program. Awarded Magna Cum Laude.

2009 – 2011 Master of Engineering, Quality Systems and Productivity — Monterrey Institute of Technology and Higher Education, Mexico

- Focus on lean manufacturing and quality management systems.

2003 – 2008 B.S., Mechatronics Engineering — Monterrey Institute of Technology and Higher Education, Mexico

- Graduated with Honors.

SKILLS

Computational Languages & Tools: MATLAB, Python, C/C++, LabVIEW, Java, ImageJ/FIJI, Icy, IMARIS, GPU programming, Parallel computing, TensorFlow, Keras, Machine Learning, R, Prism, Microsoft Excel, Zen celldiscover, IDEAS (ImageStream)

PROJECTS

2027-2028 National Natural Science Foundation of China (NSFC) Research Fund for International Excellent Young Scientists (W2632090): Applications of Biophotonics and Image Processing in Microbiology.

2025-2029 CAS Class B Strategic Priority Research Program. Project: "Construction of photosynthetic fish and nitrogen-fixing rice based on the principle of endosymbiosis". Sub-Task: Development of High-Throughput Screening Technologies.

2024 National Natural Science Foundation of China (NSFC) Special Project (32350028): Establishing cell function editing technology based on endosymbiosis.

2024-2028 National Natural Science Foundation of China (NSFC) Key Project (32330003): Three-dimensional regulatory mechanisms of cell division in multicellular cyanobacteria and their relationship with morphological diversity and peptidoglycan metabolism.

2022-2024 French National Research Agency (ANR). Labex Cell(n) Scale Groundbreaking Project. POMIMO: Polarization Microscopy for Imaging of Membrane Organization.

2019 -2022 Inria Project lab (IPL). NAVISCOPE: image-guided navigation and visualization of large data sets in live cell imaging and microscopy.

2017-2021 French National Research Agency (ANR). Project: Data Assimilation and Lattice Light Sheet imaging for endocytosis/exocytosis pathway modeling in the whole cell (DALLISH).

FELLOWSHIPS & DISTINCTIONS

- 2027–2030 Member - Mexican National Research System within the Mexican Diaspora Chairs Program (Candidato)
- 2025–2027 Elected member - Scientific Advisory Committee, GloBIAS (Global Bioimage Analysis Society)
- 2024–2027 Special Expert - President's International Fellowship Initiative (PIFI), Chinese Academy of Sciences
- 2024–2027 Young Talent - Hubei Provincial Innovative Talent Program
- 2020–2024 Member - Mexican National Research System (SNI 1)
- 2010–2014 Erasmus Mundus Fellowship for PhD
- 2008–2010 Mexican National Council of Science and Technology (CONACyT) fellowship for Master
- 2008–2010 Erasmus Mundus Fellowship for Master

PUBLICATIONS

First author publications

- [1] Senthil Kumar C. S.*, **Valades Cruz C. A.***, Sison M.*, Vesga A. G., Rey-Barroso J., Curcio V., Aleman-Castaneda L. A., Alonso M. A., Poincloux R., Mavrikis M., Brasselet S. 4polar3D single molecule imaging of 3D orientation in dense actin networks using ratiometric polarization splitting. *Nature Communications* (2026)
- [2] **Valades-Cruz C. A.***, Barth R.*, Abdellah M.*, Shaban H. A. Genome-wide analysis of the biophysical properties of chromatin and nuclear proteins in living cells with Hi-D. *Nature Protocols* (2025)
- [3] Papereux S.*, Leconte L.*, **Valades-Cruz C. A.***, Liu T., Dumont J., Chen Z., Salamero J., Kervrann C., Badoual A. DeepCristae, a CNN for the restoration of mitochondria cristae in live microscopy images. *Communications Biology* (2025)
- [4] Prigent S.*, **Valades-Cruz C.A.***, Leconte L.*, Maury L., Salamero J., Kervrann C. BioImageIT: Open-source framework for integration of image data-management with analysis. *Nature Methods* (2022)
- [5] Vaz Rimoli C.*, **Valades-Cruz C.A.***, Curcio V., Mavrikis M., Brasselet S. 4polar-STORM polarized super-resolution imaging of actin filament organization in cells. *Nature Communications* (2022)
- [6] Shaban H.*, **Valades-Cruz C. A.***, Savatier J., Brasselet S. Polarized super-resolution structural imaging inside amyloid fibrils using Thioflavine T. *Scientific Reports* (2017)
- [7] **Valades-Cruz C. A.***, Shaban H.*, Kress A., Bertaux N., Monneret S., Mavrikis M., Savatier J., Brasselet S. Quantitative nanoscale imaging of orientational order in biological filaments by polarized superresolution microscopy. *PNAS* (2016)

Corresponding author publications

- [1] Prigent S.[†], **Valades-Cruz C.A.[†]**, Leconte L., Salamero J., Kervrann C. STracking: a free and open-source python library for particle tracking and analysis. *Bioinformatics* (2022)

[†] Co-corresponding authors.

Other publications

- [1] Li S-Y., Sun Q-X., Huang D-Q., Xie T., Yu Y., He C., **Valades-Cruz C.A.**, et al. DnaK1-mediated regulation of phototactic motility in *Synechococcus elongatus* UTEX 3055. *mLife*, in press (2026).
- [2] Masson A., Prigent S., **Valades-Cruz C. A.**, Leconte L., Maury L., Salamero J., Kervrann C. BioImageIT: A novel python-based architecture for reproducible bio-image workflows. *Journal of Microscopy* (2026)
- [3] Shafaq-Zadah M., Dransart E., Hamitouche I., Wunder C., Chambon V., **Valades-Cruz C. A.**, Leconte L., Sarangi N. K., Robinson J., Bai S-K., Regmi R., Di Cicco A., Hovasse A., Bartels R., Nilsson U.J., Cianferani-Sanglier S., Leffler H., Keyes T. E., Levy D., Raunser S., Roderer D., Johannes L. Spatial N-glycan rearrangement on $\alpha 5 \beta 1$ integrin nucleates galectin-3 oligomers to determine endocytic fate. *Nature Communications* (2025)
- [4] MacDonald E., Forrester A., **Valades-Cruz C. A.**, Madsen T. D., Hetmanski J.H.R., Dransart E., Ng Y., Godbole R., Shp A. A., Leconte L., Chambon V., Ghosh D., Pinet A., Bhatia D., Lombard B., Loew D., Larsen M. R., Leffler H., Lefeber D.J., Clausen H., Blangy A., Caswell P., Shafaq-Zadah M., Mayor S., Weigert R., Wunder C., Johannes L. Growth factor-triggered de-sialylation controls glycolipid-lectin-driven endocytosis. *Nature Cell Biology* (2025)
- [5] Li S-Y., He C., **Valades-Cruz C. A.**, Zhang C-C., Yang Y. Phototactic signaling network in rod-shaped cyanobacteria: a study on *Synechococcus elongatus* UTEX 3055. *Microbiological Research* (2025)
- [6] Lemaigre C., Ceuppens A., **Valades-Cruz C. A.**, Ledoux B., Vanbeneden B., Hassan M., Zetterberg F. R., Nilsson U. J., Johannes L., Wunder C., Renard H-F., Morsomme P. N-BAR and F-BAR proteins – Endophilin-A3 and PSTPIP1 – control clathrin-independent endocytosis of L1CAM. *Traffic* (2023)
- [7] Prigent S., Nguyen H-N., Leconte L., **Valades-Cruz C. A.**, Hajj B., Salamero J., Kervrann C. SPITFIR(e): a supermaneuverable algorithm for fast denoising and deconvolution of 3D fluorescence microscopy images and videos. *Scientific Reports* (2023)
- [8] Forrester A., Rathjen S., Garcia-Castillo M. D., Bachert C., Couhert A., Tepshi L., Pichard S., Martinez J., Munier M., Sierocki R., Renard H-F., **Valades-Cruz C.A.**, Dingli F., Loew D., Lamaze C., Cintrat J.C., Linstedt A., Gillet D., Barbier J., Johannes L. Functional Dissection of the Retrograde Shiga Toxin Trafficking Inhibitor Retro-2. *Nature Chemical Biology* (2020)
- [9] Renard H-F., Tyckaert F., Lo Giudice C., Hirsch T., **Valades-Cruz C.A.**, Lemaigre C., Shafaq-Zadah M., Wunder C., Wattiez R., Johannes L., van der Bruggen P., Alsteens D., Morsomme P. Endophilin-A3 and Galectin-8 control the clathrin-independent endocytosis of CD166. *Nature Communications* (2020)
- [10] Briane V., Vimond M., **Valades-Cruz CA**, Salomon A, Wunder C, Kervrann C. A sequential algorithm to detect diffusion switching along intracellular particle trajectories. *Bioinformatics* (2020)
- [11] Salomon A., **Valades-Cruz C. A.**, Leconte L., Kervrann C. Dense Mapping of Intracellular Diffusion and Drift from Single-Particle Tracking Data Analysis, ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2020, pp. 966-970.

[12] Torrino S., Shen W., Blouin C., Kailasam Mani S., Viaris de Lesegno C., Bost P., Grassard A., Koster D., **Valades-Cruz C. A.**, Chambon V., Johannes L., Pierobon P., Soumelis V., Coirault C., Vassilopoulos S., Lamaze C. EHD2 is a mechanotransducer connecting caveolae dynamics with gene transcription. *J Cell Biol.* (2018)

[13] Banerjee A., Grazon C., Pons T., Bhatia D., **Valades-Cruz C. A.**, Johannes L., Krishnan Y., Dubertret B. A Novel Type of Quantum Dot–Transferrin Conjugate Using DNA Hybridization Mimics Intracellular Recycling of Endogenous Transferrin. *Nanoscale* (2018)

Reviews, Perspectives & Comments

[1] Ambroset M., **Valades-Cruz C. A.**, Tonfack L. B., D'Antuono R., Fuster-Barcelo C. Assessing the reproducibility of a bioimage analysis workflow characterising tissue flow in *Drosophila*. *Journal of Microscopy* (2026)

[2] **Valades-Cruz C. A.** et al. Challenges of intracellular visualization using virtual and augmented reality. *Front. Bioinform.* (2022)

[3] Johannes L., **Valades-Cruz C. A.** The final twist in endocytic membrane scission. *Nature Cell Biology* (2021)

Review assignments journals: *Nat. Communications, Bioinformatics, PLOS Comput. Biol., J. Phys. Chem. Lett. & J. Vis. Exp.*

CONFERENCES

[1] **Valades-Cruz C. A.**, Cao S., Xie T., Zhang C-C. Image analysis framework for filamentous cyanobacteria: preprocessing, segmentation and tracking. *Global BioImaging Exchange of Experience Conference*, Montreal, Canada (2025). Travel Award grantee (Chan Zuckerberg Initiative).

[2] **Valades-Cruz C. A.**, Tao J., Cao S-K., Zhang J-Y., Zhang C-C. Image analysis framework for filamentous cyanobacteria: preprocessing, segmentation, tracking and 3D volume estimation. Focus on Microscopy, Genoa, Italy (2024). Travel grant recipient (COMULISglobe).

[3] Ambroset M., **Valades-Cruz C. A.**, Tonfack L. B., D'Antuono R., Fuster-Barcelo C. Assessing the reproducibility of a bioimage analysis workflow characterising tissue flow in *Drosophila*. *GloBIAS In-Person Workshop*, Gothenburg, Sweden (2024). Travel grant recipient (Chan Zuckerberg Initiative).

[4] **Valades-Cruz C. A.** Advanced microscopy and image processing for biological applications. *Global BioImaging Exchange of Experience Conference*, University of Cape Town, South Africa (2023). Travel Award grantee (Chan Zuckerberg Initiative).

[5] **Valades-Cruz C. A.** Data management and analysis tools. *Inserm Workshop 268 – Light Sheet Fluorescence Microscopy: Principles and Systems Overview*, Bordeaux, France (2022).

[6] **Valades-Cruz C. A.**, Leconte L., Wunder C., Kervrann C., Johannes L., Salamero J. 3D tracking of endocytic and exocytic events using lattice light sheet microscopy. *QBI 2020 – Quantitative BioImaging Conference*, Oxford, United Kingdom (2020).

[7] **Valades-Cruz C. A.**, Forrester A., Wunder C., Kervrann C., Salamero J., Johannes L. 3D tracking of endocytic events using lattice light sheet microscopy. *QBI 2019 – Quantitative BioImaging Conference*, Rennes, France (2019).